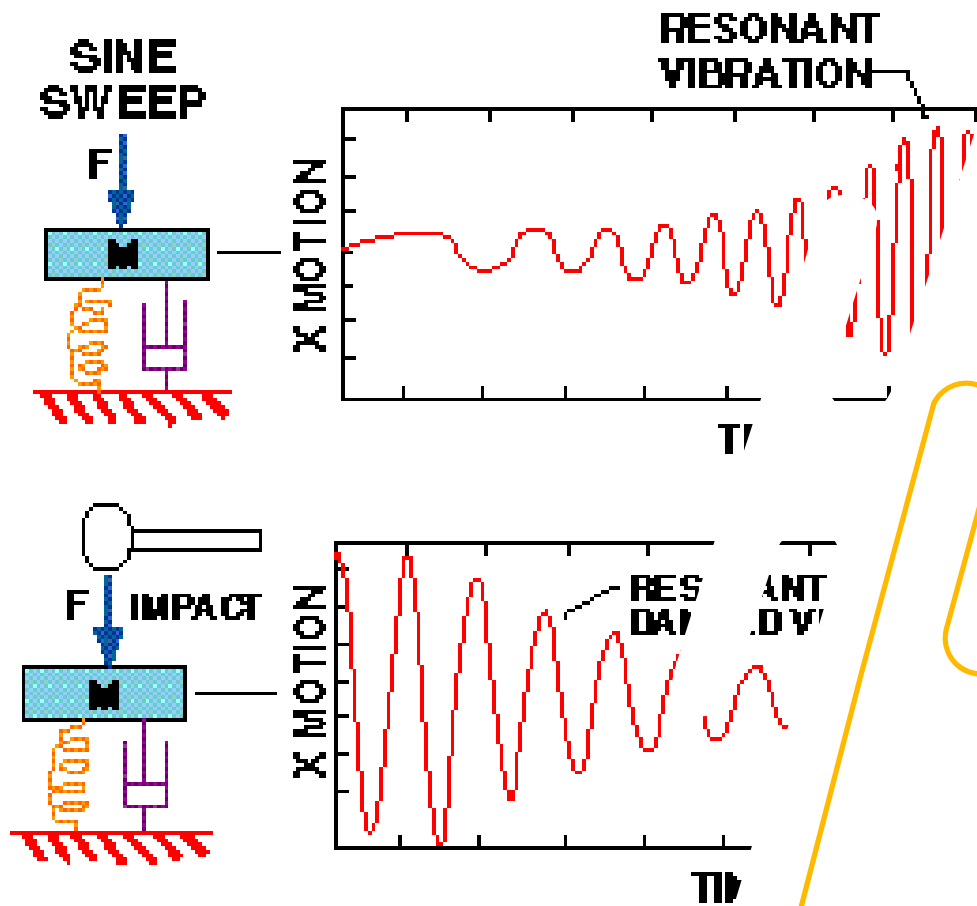




Fundamental of System Vibration

MENG 4050

REZA MADJLESI

Why Theory?

The understanding of vibration theory is important for the professionals involved in vibration measurements and vibration-based condition monitoring (VCM).

Understanding the dynamics of a simple system such as a spring and mass system is always required. This concept can be utilized to understand the dynamics and vibration behavior of a complex system.

Ref: Industrial Approaches in VBC Monitoring, Jyoti Sinha

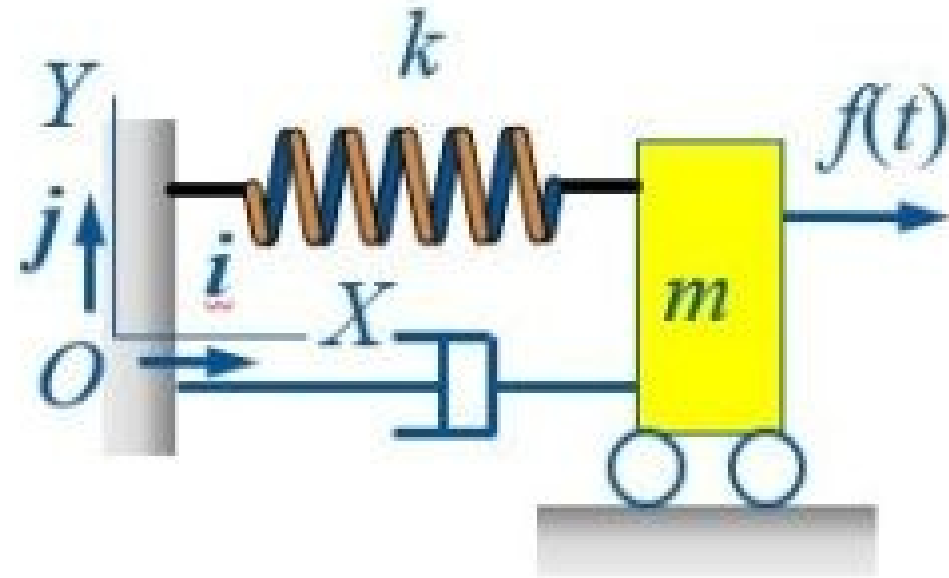


1DOF System

A simple system consists of a spring of stiffness (k) and a mass (m)

When the system is disturbed from rest, it oscillates about the static equilibrium position. This oscillation is the natural oscillation of the mass within the spring-mass system without any external force.

The related frequency is the natural frequency of spring and mass system



Machine Vibration Health Monitoring, Theory Background

- 1. SDOF Equation of Motion, Undamped
- 2. SDOF Equation of Motion, Damped
- 3. Critically Damped System
- 4. Over Damped System
- 5. Under Damped System
- 6. Forced Vibration
- 7. Concept of Mode Shapes
- 8. Rotor Dynamics and Bearings





Section One

We will cover SDOF Free Vibration

- Undamped vs. Damped
- Under Damped, Over Damped, Critically Damped Systems

SDOF Equation of Motion, Undamped

$$M\ddot{Y}(t) + kY(t) = 0$$

$$-M\omega_n^2 + kY(t) = 0$$

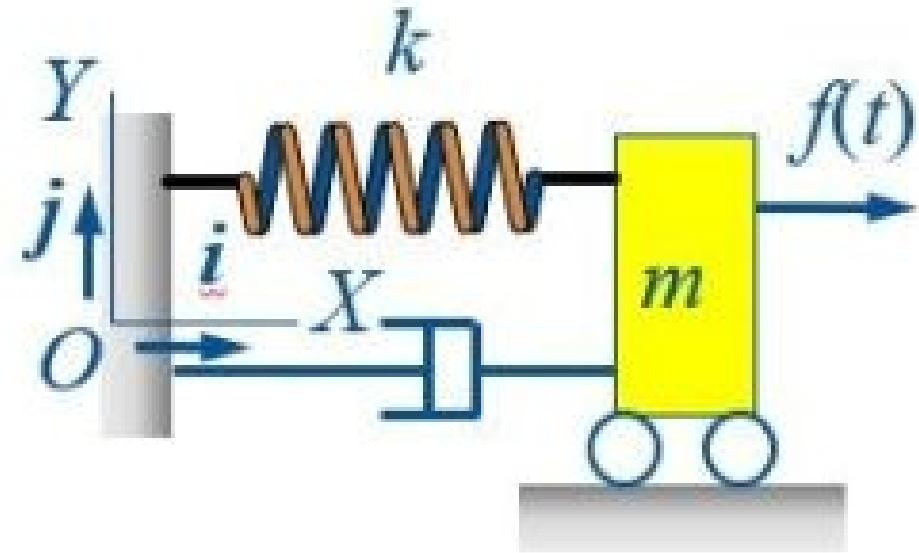
$$\omega_n = \sqrt{\frac{k}{M}}$$

$$T = \frac{2\pi}{\omega_n}$$

$$f_n = \frac{1}{T} = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{M}}$$

$$Y(t) = A \sin \omega_n t + B \cos \omega_n t$$

$$Y(t) = \frac{Y(\dot{0})}{\omega_n} \sin \omega_n t + Y(0) \cos \omega_n t$$



SDOF Equation of Motion, Damped

$$M\ddot{Y}(t) + C\dot{Y}(t) + kY(t) = 0$$

$$(MS^2 + CS + k)Y(0)e^{St} = 0$$

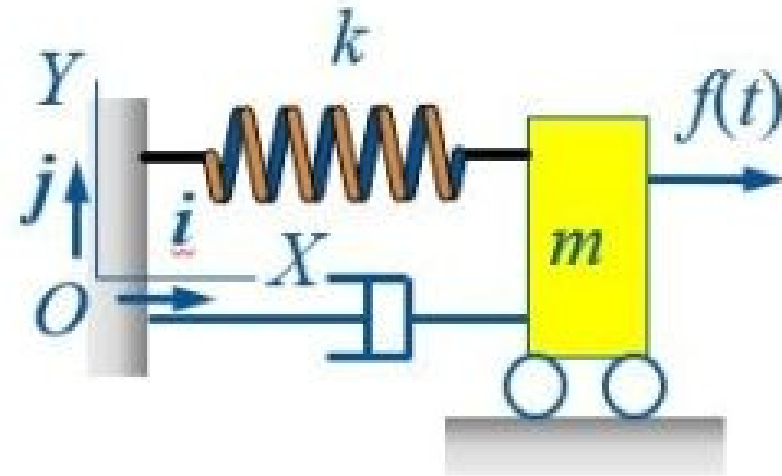
$$\omega_n = \sqrt{\frac{k}{M}}$$

$$T = \frac{2\pi}{\omega_n}$$

$$f_n = \frac{1}{T} = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{M}}$$

$$S^2 + \frac{C}{M}S + \frac{k}{M} = 0$$

$$S_{1,2} = -\frac{c}{2M} \pm \sqrt{\left(\frac{c}{2M}\right)^2 - \frac{K}{M}} Y(t) = Ae^{s_1 t} + Be^{s_2 t}$$



Critically Damped System

$$DF = \sqrt{\left(\frac{C}{2M}\right)^2 - \frac{K}{M}}, \text{ Discriminant Factor}$$

$$DF = 0, C = 2M \sqrt{\frac{K}{M}} = 2M\omega_n = 2\sqrt{KM}$$

$$Y(t) = (A + Bt)e^{St},$$

$$s = -\frac{c}{2m} = -\frac{c}{c_c} \frac{c_c}{2m} = -\zeta \frac{2m\omega_n}{2m} = -\zeta\omega_n$$

$$y(t) = (y(0) + (\dot{y}(0) + \omega_n y(0))t) e^{-\omega_n t}$$

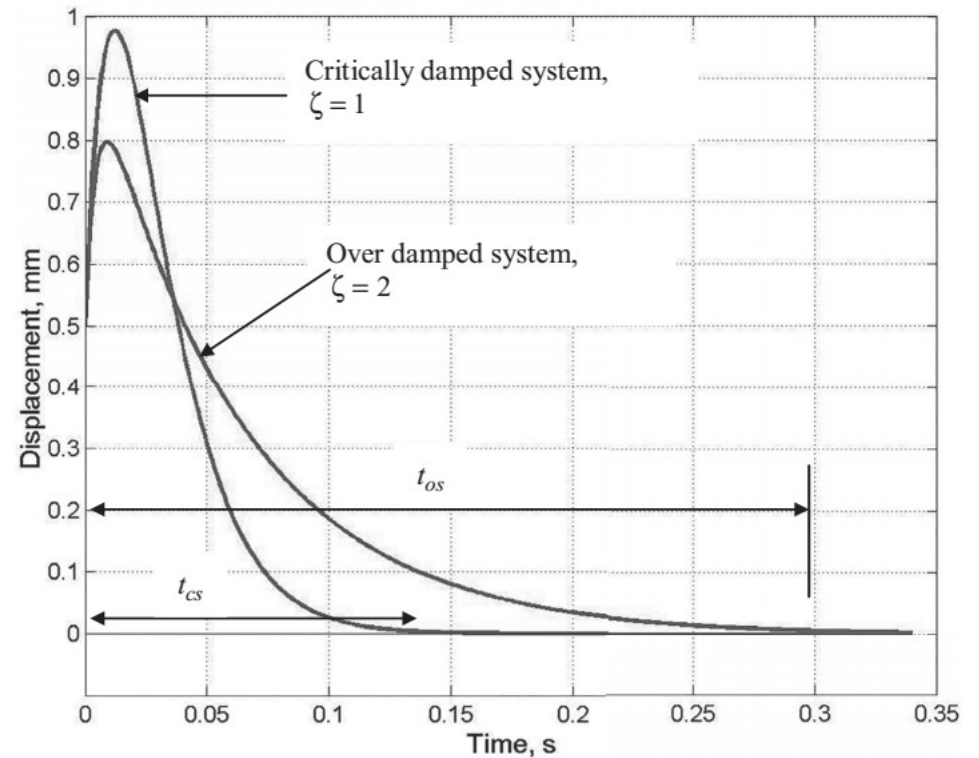


FIGURE 2.4 Comparison of dynamic behavior between an over damped and a critically damped systems.

Over Damped System

$$DF = \sqrt{\left(\frac{C}{2M}\right)^2 - \frac{K}{M}} > 0, C > C_c, \zeta > 1$$

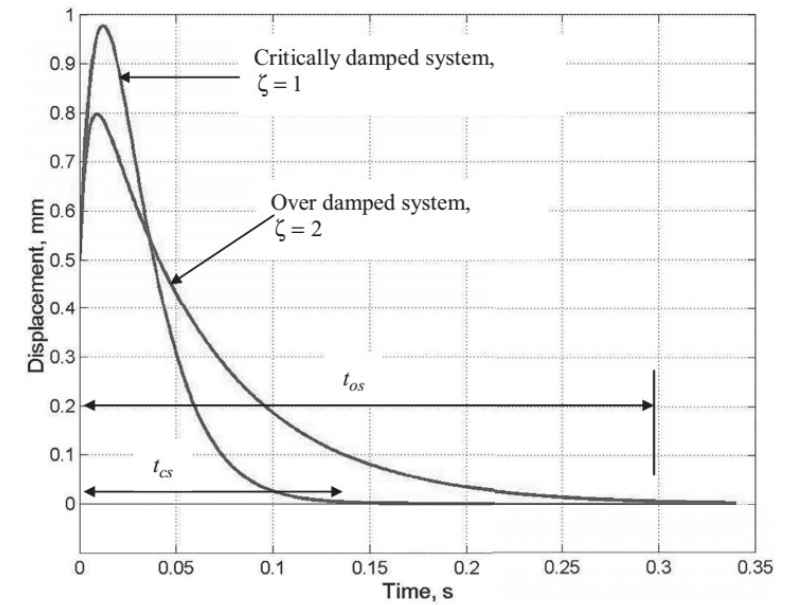


FIGURE 2.4 Comparison of dynamic behavior between an over damped and a critically damped systems.

$$y(t) = Ae^{\left(-\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n t} + Be^{\left(-\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n t}$$

where

$$A = \frac{\dot{y}(0) + \left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n y(0)}{2\omega_n \sqrt{\zeta^2 - 1}}, \text{ and } B = \frac{-\dot{y}(0) - \left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n y(0)}{2\omega_n \sqrt{\zeta^2 - 1}}$$

Under Damped System

$$\Rightarrow s_{1,2} = -\zeta\omega_n \pm j\sqrt{1-\zeta^2}\omega_n$$

$$s_{1,2} = \lambda_{1,2} = \pm j\sqrt{1-\zeta^2}\omega_n = -\zeta\omega_n \pm j\omega_d$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$y(t) = e^{-\zeta\omega_n t} \left(A e^{j\sqrt{1-\zeta^2}\omega_n t} + B e^{-j\sqrt{1-\zeta^2}\omega_n t} \right)$$

$$\Rightarrow y(t) = e^{-\zeta\omega_n t} \left(A e^{j\omega_d t} + B e^{-j\omega_d t} \right)$$

$$y(t) = e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

$$\text{where } C_1 = y(0), C_2 = \frac{\dot{y}(0) + \zeta\omega_n y(0)}{\omega_d}$$

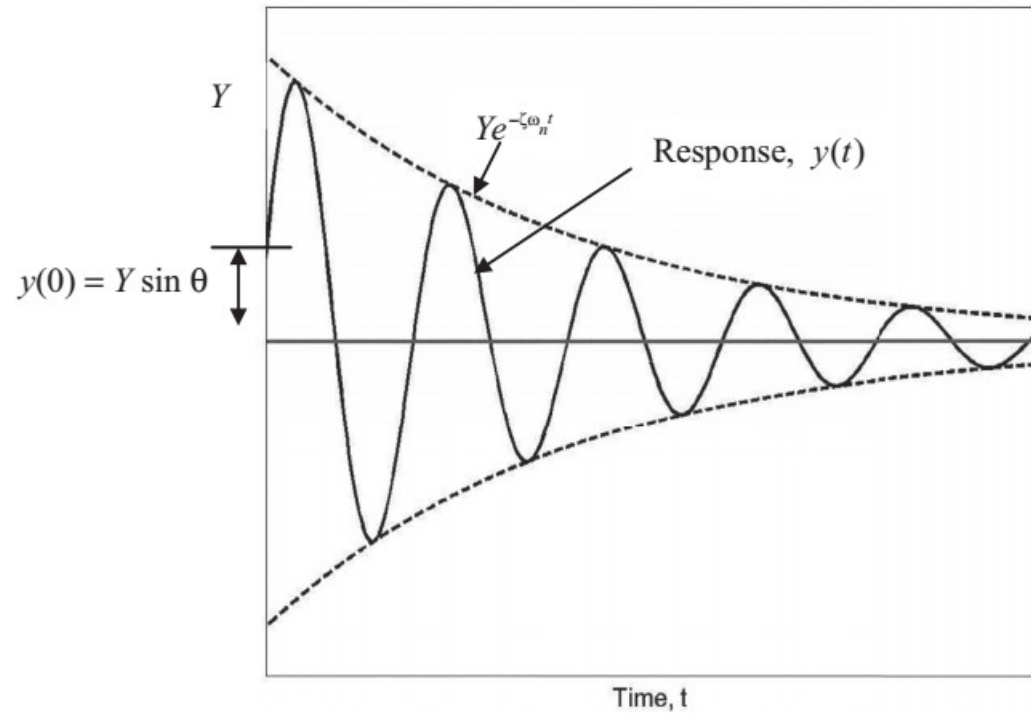


FIGURE 2.6 Response of the under damped SDOF system.



Section Two

We will cover SDOF Forced Vibration

- Forced Vibration

Forced Vibration

$$M\ddot{Y}(t) + C\dot{Y}(t) + kY(t) = f(t)$$

$$(-M\omega^2 + jC\omega + k)Ye^{i\omega t} = F_0 e^{i\omega t}$$

$$Y(t) = Y_c(t) + Y_p(t)$$

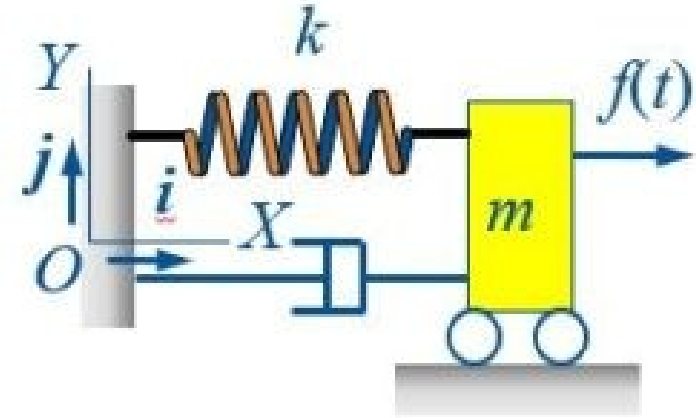
$$Y_p(t) = \frac{\frac{F_0}{k}}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right) + j\left(2\zeta\frac{\omega}{\omega_n}\right)} e^{i\omega t}$$

$$Y_p(t) = \frac{\frac{F_0}{k}}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} e^{i\omega t}$$

$$r = \frac{\omega}{\omega_n}, \quad \phi = \tan^{-1} \frac{(2\zeta r)}{(1 - r^2)}$$

$$Y_p(t) = \frac{F_0}{k} D e^{i(\omega t - \phi)},$$

$$D = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \text{ amplification or } \\ \text{magnification factor}$$

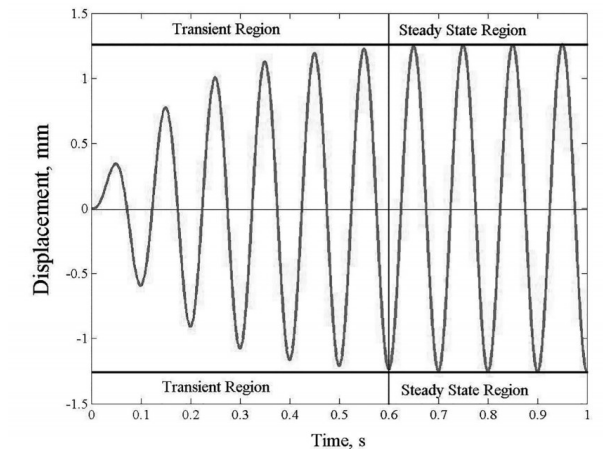
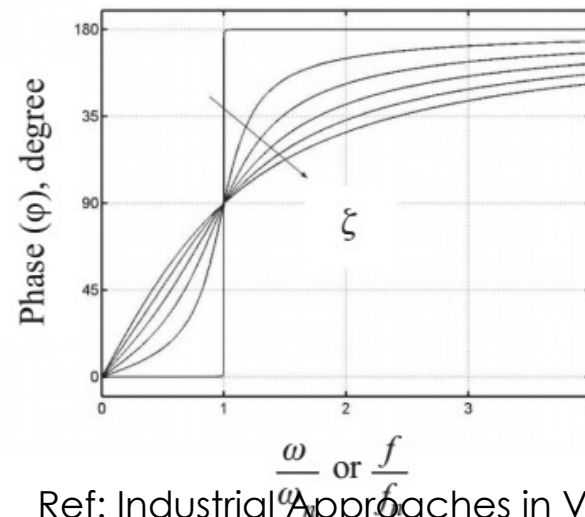
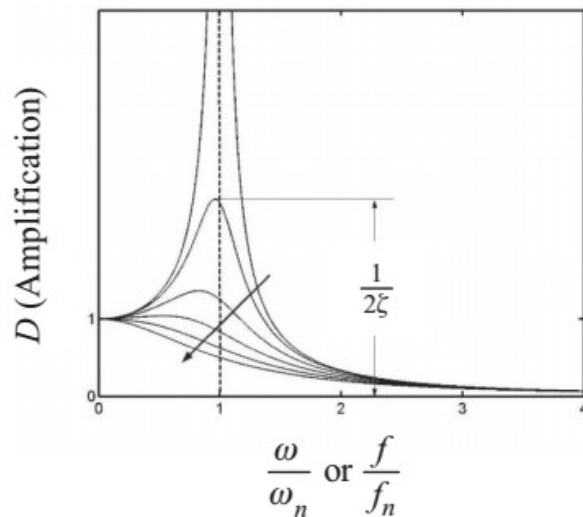
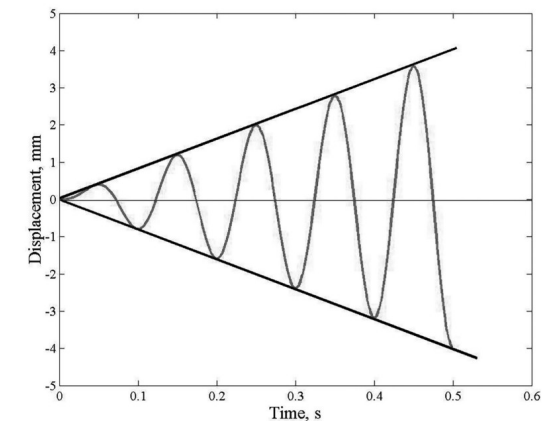
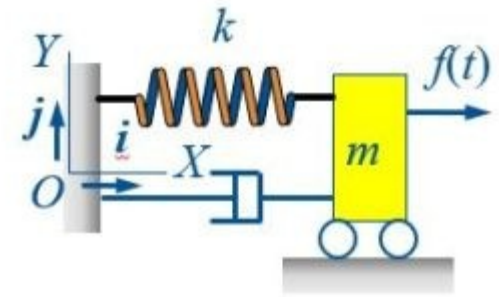


Forced Vibration

$$r = \frac{\omega}{\omega_n}, \quad \phi = \tan^{-1} \frac{(2\zeta r)}{(1-r^2)}$$

$$Y_p(t) = \frac{F_0}{k} D e^{i(\omega t - \phi)}, \quad D = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \text{ amplification or}$$

magnification factor, $r = 1, D = \frac{1}{2\zeta}$. Machine speed should not be in $\pm 20\%$ of natural frequency, $\text{RPM} = 60 \times F$



Ref: Industrial Approaches in VBC Monitoring, Jyoti Sinha



Section Three

We will cover Applied
Vibration Concepts

- Mode Shapes
- Rotor Dynamics
- Bearing Vibration
- Unbalanced Response
- Critical Speed
- Machine Faults

Concept of Mode Shapes

When any system is vibrating at a natural frequency then each point within the system is going to oscillate at natural frequency but the amplitude and direction of deformation of each point may or may not be same.

This typical shape of deformation of the system at a natural frequency is called mode shape for that natural frequency.

Modal shape	Natural frequency
First mode 	$\omega_1 = (1.875)^2 \sqrt{\frac{EI}{mL^4}}$ $f_1 = 6,19 \text{ Hz}$
Second mode 	$\omega_2 = (4.694)^2 \sqrt{\frac{EI}{mL^4}}$ $f_2 = 38,79 \text{ Hz}$
Third mode 	$\omega_3 = (7.855)^2 \sqrt{\frac{EI}{mL^4}}$ $f_3 = 108,60 \text{ Hz}$

Machine Vibration Rotor Dynamics

During the machine operation (i.e., shaft rotation) the rotor likely to be deformed in both vertical and horizontal directions

it is normal practice to show the mode shapes of different modes in both axes simultaneously to trace the path of rotor deformation during the machine vibration.

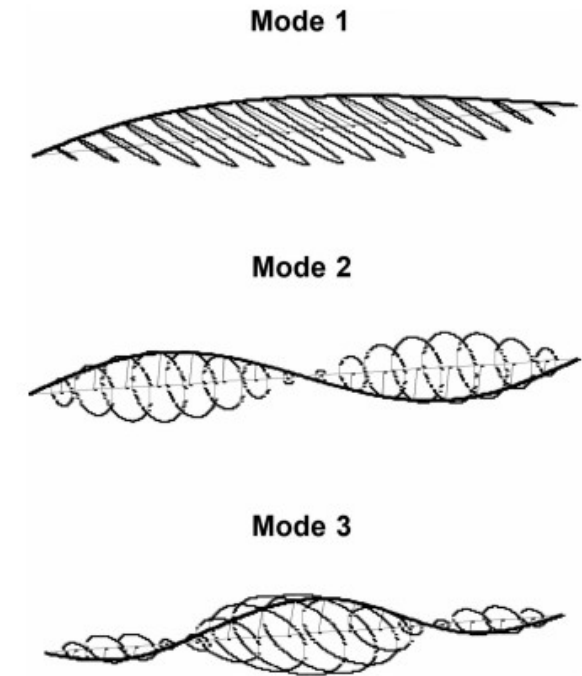


FIGURE 2.17 Modeshapes of a rotor.

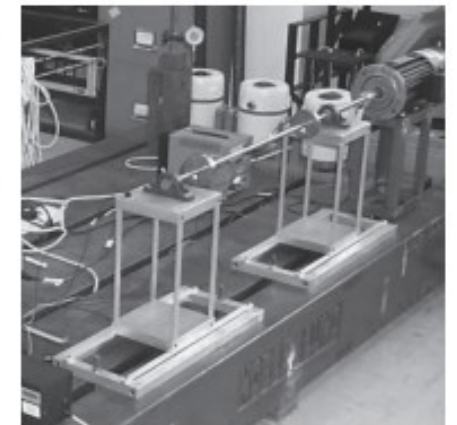
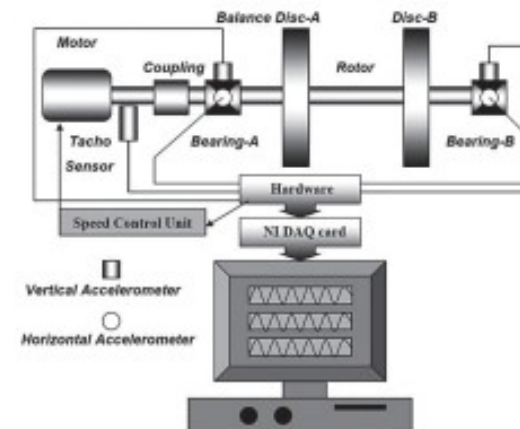


FIGURE 2.16 An experimental rig with vibration instruments—schematic and photograph.

Bearing Vibration

Why does the machine vibrate when operating?

Why do the vibration amplitudes show a decreasing trend in general from high rotating speed to low rotating speed?

Why are there a few vibration amplifications during machine rundown?

Run down from 2500 to 300 RPM

Ref: Industrial Approaches in VBC Monitoring, Jyoti Sinha

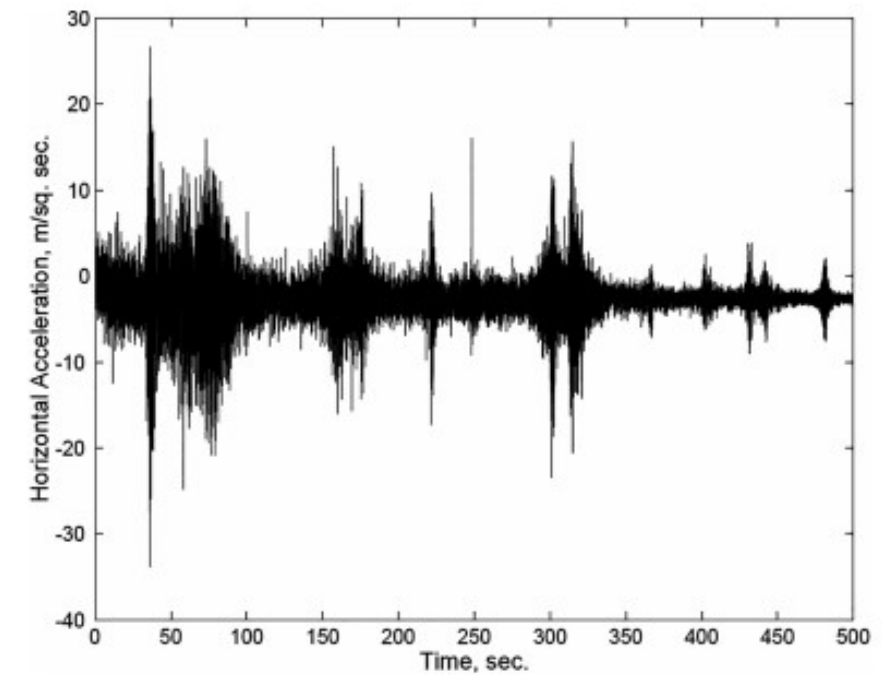
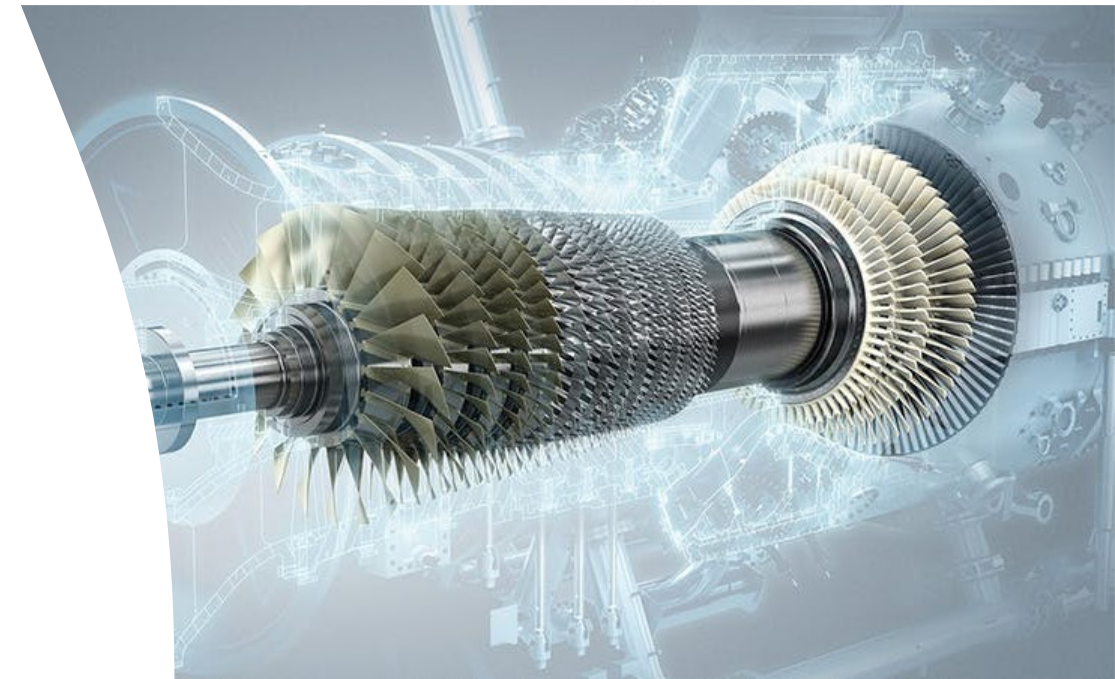


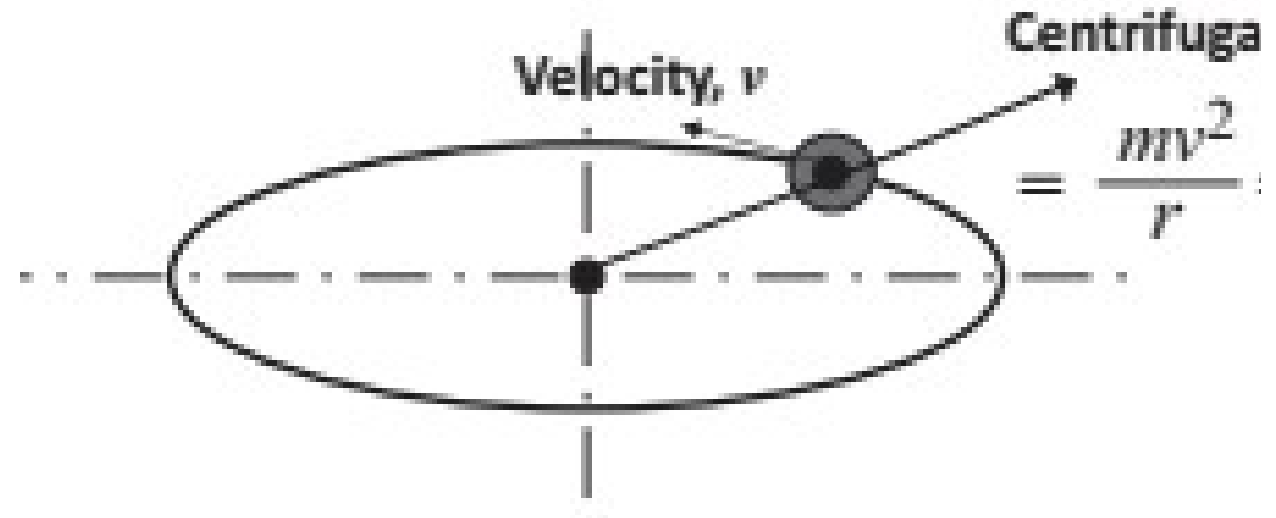
FIGURE 2.18 Vibration acceleration at Bearing A in horizontal direction.



Machine Vibration Unbalanced Responses

The rotor unbalance force is originated from the concept of the centrifugal force acting on an object during circular motion.

$F_{\text{Centrifugal}} = mv^2/r = mr\omega^2$, where $v = r\omega$,
 ω is the angular velocity, radian/s
(rad/s).



Critical Speeds

The amplification of vibration during machine runup or rundown indicates that the machine speed is passing through the machine natural frequency.

Machine natural frequency is known as the critical speed of the machine.

Machines may have several natural frequencies (or critical speeds) in the speed range of zero speed to the machine operating speed.

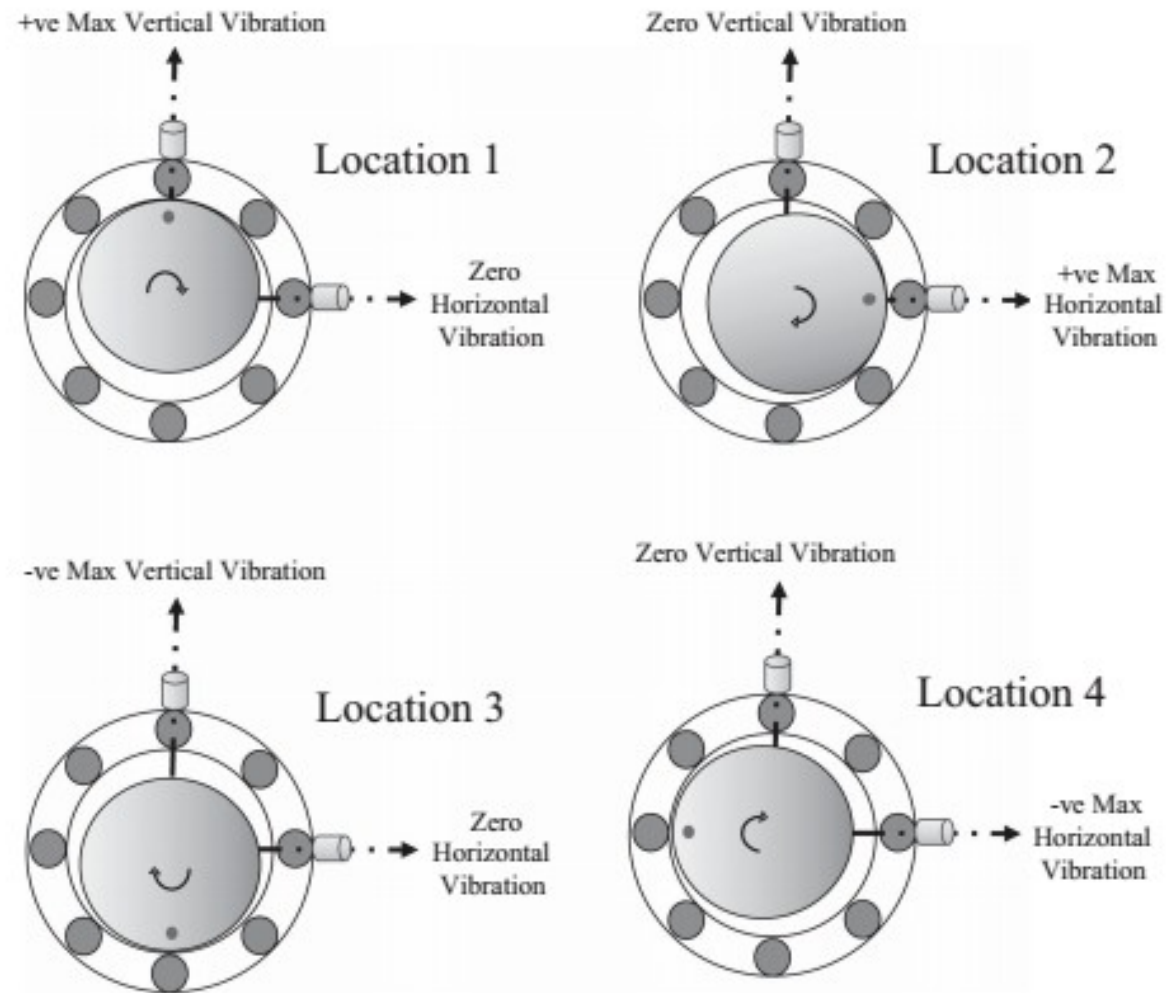


FIGURE 2.22 Typical machine vibration mechanisms under the rotor unbalance only.

Machine Vibration Machine Faults

Mechanical looseness, rotor crack may:

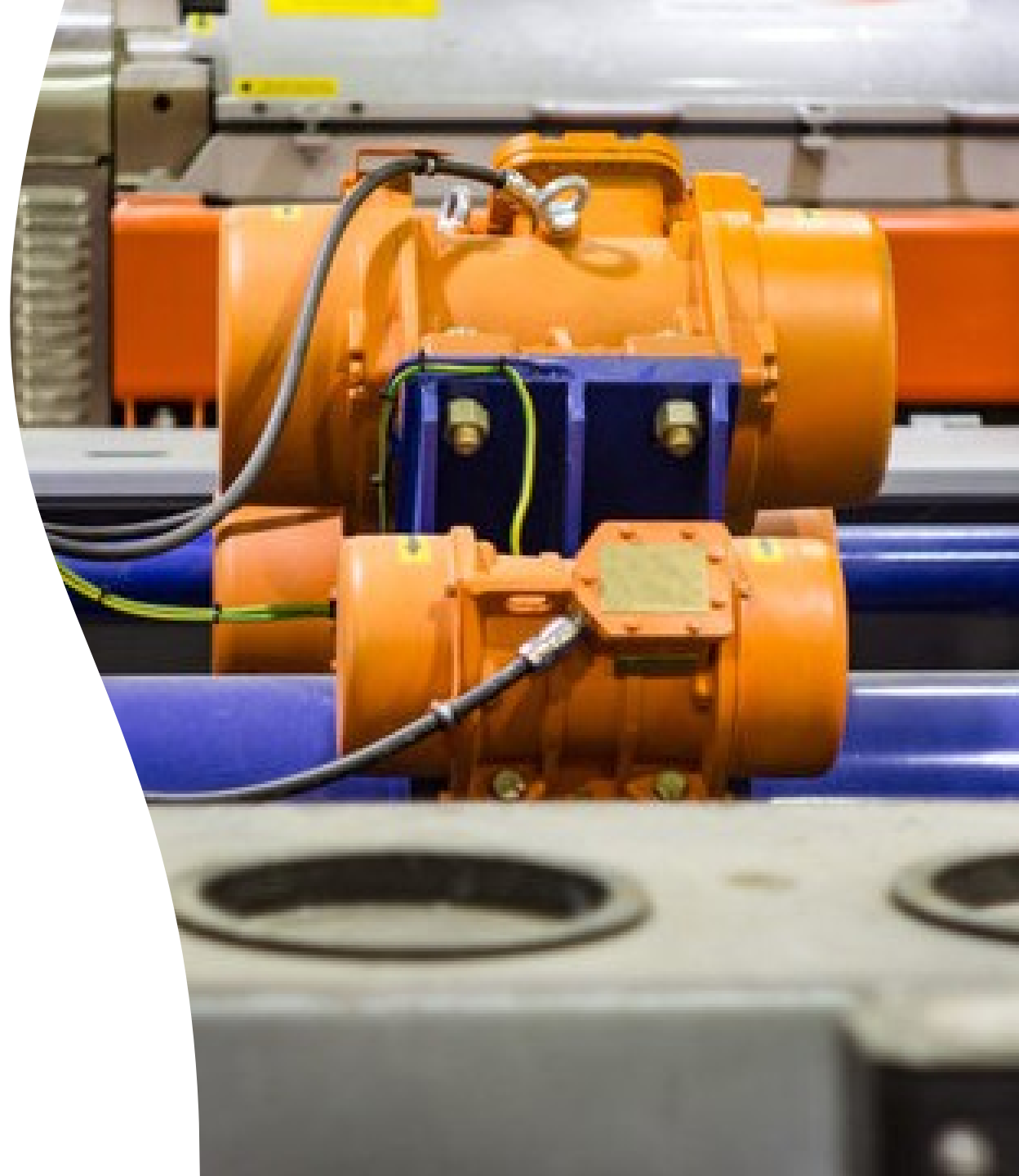
1. generate a defect related force in addition to the existing unbalance force
2. time dependent change in the system stiffness

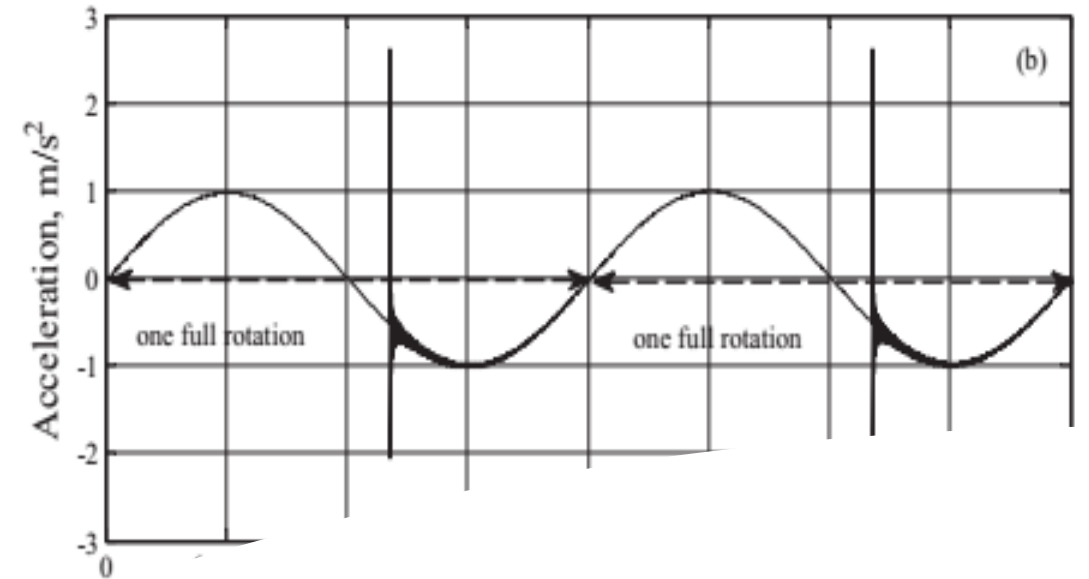
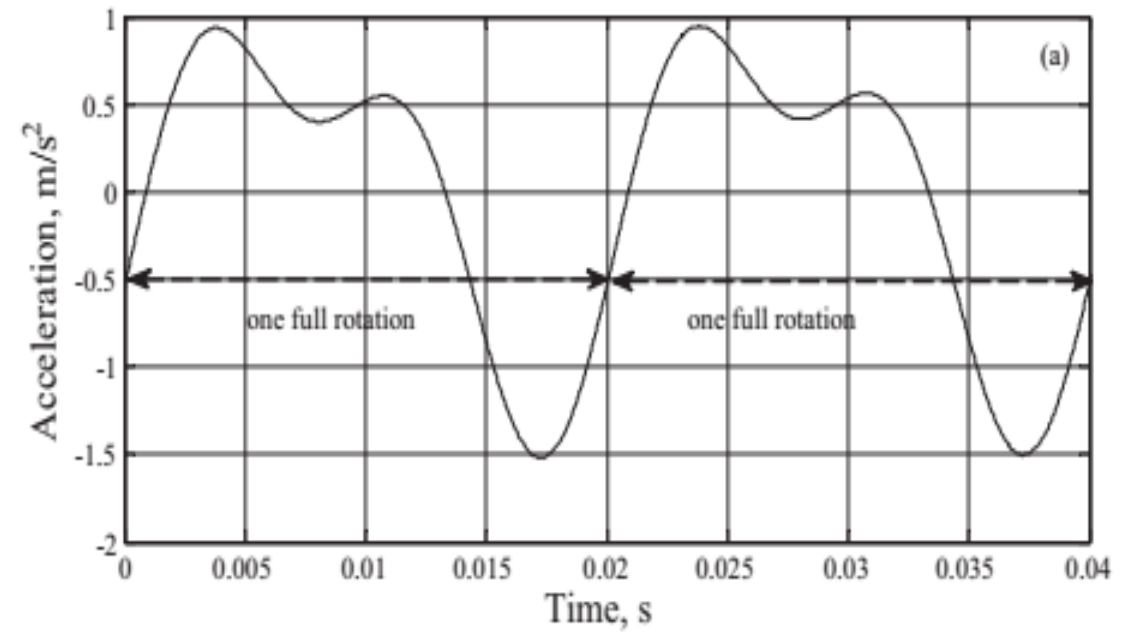
The vibration response is not pure sinusoidal response. It may change to periodic response with the fundamental frequency related to the machine RPM

- A defect in a ball bearing is going to generate an impulsive signal during each shaft rotation.

the vibration response of machine is to contain both rotor unbalance response plus response due to bearing defect

Ref: Industrial Approaches in VBC Monitoring, Jyoti Sinha





Machine Vibration Machine Faults



Thank you

Office hours:

Please send all questions through course message